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DEVICE PROVIDING MULTIPLE THRESHOLD VOLTAGES AND METHODS OF MAKING THE SAME

Abstract

A method includes receiving a structure including a first region and a second region, forming a dielectric layer over the first region and the second region, forming a first patterned layer of a first dipole material on the dielectric layer in the first region, performing a first thermal drive-in operation to drive the first dipole material into the dielectric layer, forming a second patterned layer of a second dipole material on the dielectric layer in the second region, performing a second thermal drive-in operation to drive the second dipole material into the dielectric layer, performing a thermal operation to adjust distribution of the first dipole material or both the first and the second dipole materials in the dielectric layer, and forming a gate electrode layer over the dielectric layer. A portion of the first region overlaps with the second region.

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Background/Summary

PRIORITY DATA [0001] This is a continuation application of U.S. patent application Ser. No. 18/738,303, filed on Jun. 10, 2024, entitled “Device Providing Multiple Threshold Voltages and Methods of Making the Same”, which is a continuation application of U.S. patent application Ser. No. 17/464,091, filed on Sep. 1, 2021, now issued as U.S. Pat. No. 12,009,400, entitled “Device Providing Multiple Threshold Voltages and Methods of Making the Same”, which is a non-provisional application of and claims priority to U.S. Provisional Patent Application No. 63/149,315, filed on Feb. 14, 2021, entitled “Device Providing Multiple Threshold Voltages and Methods of Making the Same”, each of which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of IC processing and manufacturing, and for these advancements to be realized, similar developments in IC processing and manufacturing are needed.

[0003] For example, nanosheet-based devices have been introduced in an effort to improve gate control by increasing gate-channel coupling, reduce OFF-state current, and reduce short-channel effects (SCEs). Nanosheet-based devices include a plurality of channel layers stacked together to form the transistor channels which are engaged by a gate structure. The nanosheet-based devices are compatible with conventional complementary metal-oxide-semiconductor (CMOS) processes, allowing them to be aggressively scaled down while maintaining gate control and mitigating SCEs. However, due to the complex device structures and reduced spacing between features, it may be challenging to accomplish certain functions, such as to provide multiple threshold voltages, without incurring penalty to other performance characteristics. Therefore, although conventional technologies have been generally adequate for their intended purposes, they are not satisfactory in every respect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIGS. **1** and **14** are flow charts illustrating methods for fabricating devices of the present disclosure according to some embodiments of the present disclosure.

[0006] FIGS. **2**A and **15** are plan views of embodiments of devices of the present disclosure according to some embodiments of the present disclosure.

[0007] FIG. **2**B is a three-dimensional (3D) perspective view of a nanosheet-based transistor of an embodiment of a device of the present disclosure constructed according to some embodiments of the present disclosure.

[0008] FIG. **2**C is a cross-sectional view of the nanosheet-based transistor of FIG. **2**B along the line A-A′ according to some embodiments of the present disclosure.

[0009] FIGS. **3**, **4**, **5**, **6**, **7**, **8**, **9**, **10**, **11**, **12**, **16**, **17**, **18**, **19**, **20**, and **21** are cross-sectional views of an embodiment of a device of the present disclosure, or portions thereof, constructed at various fabrication stages according to some embodiments of the present disclosure.

[0010] FIGS. **13**, **22**, **23**A, **23**B, **23**C, **24**, **25**A, **25**B, **26**A, and **26**B are data illustrating various aspects of embodiments of the present disclosure.

DETAILED DESCRIPTION

[0011] The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0012] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0013] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Still further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within +/−10% of the number described, unless otherwise specified. For example, the term “about 5 nm” encompasses the dimension range from 4.5 nm to 5.5 nm.

[0014] The present disclosure is generally related to ICs and semiconductor devices and methods of forming the same. More particularly, the present disclosure is related to semiconductor devices having multiple threshold voltages (Vt) (hereinafter referred to as multi-Vt devices). As the advanced technology nodes continue to scale down, it has become increasingly challenging to develop such multi-Vt devices. Typically, various additional material layers may be required in order to engineer the devices to provide multiple threshold voltages. These material layers occupy

certain spaces (and/or volumes) on the semiconductor chips which impedes the effort of scale-down. In some examples, the devices may not have sufficient space to accommodate such additional material layers. For example, nanosheet-based devices (sometimes referred to as gate-all-around (GAA) devices, multi-bridge-channel (MBC) devices, or other similar names) include a plurality of channel layers stacked one on top of another. The gate stacks are formed in the very narrow spacing between vertically adjacent channel layers, where additional material layers are sometimes impractical to reliably form. Moreover, volumes of these additional material layers may further adversely affect device performances, such as leading to a penalty to the channel resistance (R.sub.ch). Accordingly, the present disclosure provides processes and methods that allows formation of multi-Vt devices without the volume requirement. The devices presented herein may be a complementary metal-oxide-semiconductor (CMOS) device, a p-type metal-oxide-semiconductor (PMOS) device, or an n-type metal-oxide-semiconductor (NMOS) device. One of ordinary skill may recognize other examples of semiconductor devices that may benefit from aspects of the present disclosure. Moreover, although the disclosure uses nanosheet-based devices as an example, one of ordinary skill may recognize other examples of semiconductor devices that may benefit from aspects of the present disclosure. For example, other types of metal-oxide semiconductor field effect transistors (MOSFETs), such as planar MOSFETs, FinFETs, other multi-gate FETs may benefit from aspects of the present disclosure.
[0015] FIG. **1** is a flow chart of an embodiment of a method **1000** of the present disclosure for preparing an embodiment of a multi-Vt device **10** (or simply device **10**) of the present disclosure. FIG. **2**A is a plan view of the device **10** according to an embodiment of the present disclosure. FIG. **2**B is a three-dimensional (3D) perspective view of a nanosheet-based transistor **100**, which is representative of a component of the device **10** of FIG. **2**A according to some embodiments of the present disclosure. FIG. **2**C is a cross-sectional view of the nanosheet-based transistor **100** of FIG. **2**B along the line A-A′ according to some embodiments of the present disclosure. FIGS. **3-12** are cross sectional views or expanded cross-sectional views of the device **10** (or portions thereof) at various fabrication stages according to embodiments of the present disclosure.
[0016] Referring to block **1010** of FIG. **1** and to FIGS. **2**A-**2**C, an example nanosheet-based multi-Vt device **10** is received. The device **10** includes a plurality of nanosheet-based transistors (or simply transistors), such as transistors **100**A-**100**D and **100**A′-**100**D′. In the depicted embodiments, the transistors **100**A and **100**A′ are formed in a substrate region **102**A of a semiconductor substrate **102**; the transistors **100**B and **100**B′ are formed are formed in a substrate region **102**B of a semiconductor substrate **102**; the transistors **100**C and **100**C′ are formed are formed in a substrate region **102**C of a semiconductor substrate **102**; and the transistors **100**D and **100**D′ are formed are formed in a substrate region **102**D of a semiconductor substrate **102**. Moreover, in the depicted embodiments, transistors **100**A-**100**D may be configured as n-type transistors, while the transistors **100**A′-**100**D′ may be configured as p-type transistors. In some embodiments, the transistors **100**A-**100**D each have a different threshold voltage; and the transistors **100**A′-**100**D′ each have a different threshold voltage. Accordingly, the device **10** provides n-type transistors and p-type transistors each offering four (4) different threshold voltages. As described later, these total of eight (8) threshold voltages may be achieved by configuring gate dielectric layers differently with respect to dipole elements. The transistors **100**A-**100**D and **100**A′-**100**D′ may include similar or different device structures. In the depicted embodiments, transistors **100**A-**100**D and **100**A′-**100**D′ are each nanosheet-based transistors and have similar device structures, such as the device structure of transistor **100** illustrated in FIGS. **2**B and **2**C. FIGS. **2**A-**2**C have been abbreviated to provide a general picture of the device **10**, and do not include all details. For example, shapes, sizes, and relative positions of the transistors shown in FIGS. **2**A-**2**C have been simplified and/or conceptualized, and are not intended to be limiting. Additional details are described in conjunction with subsequent figures.
[0017] Referring to FIGS. **2**B and **2**C, nanosheet-based transistor **100** (or simply transistor **100**)

may be representative of one or more of the transistors **100**A-**100**D and **100**A′-**100**D′ of FIG. **2**A. In other words, FIGS. **2**B and **2**C illustrates a portion of FIG. **2**A. As illustrated, the transistor **100** includes a semiconductor substrate **102** (or simply substrate **102**). The substrate **102** contains a semiconductor material, such as bulk silicon (Si), germanium (Ge), silicon germanium (SiGe), silicon carbide (SiC), gallium arsenic (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), and/or indium antimonide (InSb), or combinations thereof. The substrate **102** may also include a semiconductor-on-insulator substrate, such as Si-on-insulator (SOI), SiGe-on-insulator (SGOI), Ge-on-insulator (GOI) substrates. Fin structures (or fins) **104** are formed over the substrate **102**, each extending lengthwise horizontally in an X-direction and separated from each other horizontally in a Y-direction. The X-direction and the Y-direction are perpendicular to each other, and the Z-direction is orthogonal (or normal) to a horizontal XY plane defined by the X-direction and the Y-direction. The substrate **102** may have its top surface parallel to the XY plane. As described above, the substrate **102** include substrate regions **102**A-**102**D.

[0018] The fin structures **104** each have a source region **104***a* and a drain region **104***a* disposed along the X-direction. The source region **104***a* and the drain region **104***a* are collectively referred to as the source/drain regions **104***a*. Epitaxial source/drain features **500** are formed in or on the source/drain regions **104***a* of the fin structure **104**. In some embodiments, the epitaxial source/drain features **500** are configured to be part of the PMOS transistor. Accordingly, the epitaxial source/drain features **500** may include any suitable p-type semiconductor materials, such as Si, SiGe, Ge, SiGeC, or combinations thereof. In some embodiments, the epitaxial source/drain features **500** are configured to be part of the NMOS transistor. Accordingly, the epitaxial source/drain features **500** may include any suitable n-type semiconductor materials, such as Si. The epitaxial source/drain features **500** may further be doped in-situ or ex-situ. For example, the epitaxially grown SiGe source/drain features of a PMOS may be doped with boron (B) to form Si:Ge:B source/drain features; and the epitaxially grown Si source/drain features of an NMOS may be doped with carbon to form silicon:carbon (Si:C) source/drain features, doped with phosphorous to form silicon:phosphorous (Si:P) source/drain features, or both carbon and phosphorous to form silicon carbon phosphorous (Si:C:P) source/drain features. Multiple processes, including etching and growth processes (such as epitaxial processes), may be employed to grow the epitaxial source/drain features **500**. One or more annealing processes may be performed to activate the dopants in the epitaxial source/drain features **500**. In some embodiments, the epitaxial source/drain features **500** may merge together, for example, along the Y-direction between adjacent fin structures **104** to provide a larger lateral width than an individual epitaxial source/drain feature.

[0019] The fin structures **104** each further have a channel region **104***b* disposed between and connecting the source/drain regions **104***a*. The fin structures **104** each include a stack of channel layers **120** (also interchangeably referred to as "semiconductor layers **120**," "suspended semiconductor layers **120**," or "suspended channel layers **120**"). The stack of channel layers **120** occupy the channel region **104***b* of the fin structures **104** and extends vertically (e.g. along the Z-direction) from the substrate **102**. Each of the channel layers **120** connects a pair of epitaxial source/drain features **500**. The channel layers **120** may each be in one of many different shapes, such as wire (or nanowire), sheet (or nanosheet), bar (or nano-bar), and/or other suitable shapes, and may be spaced away from each other. In the depicted embodiments, there are three channel layers **120** in the stack. However, there may be any appropriate number of layers in the stack, such as 2 to 10 layers. In some embodiments, the channel layers **120** are on the nanometer scale (e.g. having at least one dimension that is about 1 nm to about 100 nm). Accordingly, the channel layers **120** are referred to as nanostructures, and the transistors are referred to as nanostructure (or nanosheet) based transistors.

[0020] In some embodiments, the fin structures **104** may be formed by first forming a stack of layers over the substrate **102**. The stack of layers may include semiconductor layers **110** and the semiconductor layers **120** alternating with each other. The material compositions of the semiconductor layers **110** and semiconductor layers **120** are configured such that they have an etching selectivity in a subsequent etching process. For example, in some embodiments, the semiconductor layers **110** contain silicon germanium (SiGe), while the semiconductor layers **120** contain silicon (Si). The stacks of layers (and in some embodiments, the substrate portion therebeneath) are then collectively patterned into the fin structures **104** such that the fin structure **104** each extend lengthwise along the X-direction. The patterning may be by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the fins. The patterning may utilize multiple etching processes which may include a dry etching and/or wet etching. The fin structures **104** may have lateral widths along the Y-direction that are the same between each other or different from each other.

[0021] The semiconductor layers **110** are subsequently removed, thus are also referred to as the sacrificial semiconductor layers **110**. Meanwhile, the patterned semiconductor layers **120** later serve as the channel for transistors, thus are also referred to as the channel layers **120**. The channel layers **120** may each engage with a single gate structure **250**. The gate structure **250** includes gate dielectric layer **246** and gate electrode layer **248**. In the depicted embodiments, the gate structure **250** further includes interfacial layer **242**. However, in some other embodiments, the interfacial layer **242** may be omitted. Note that the gate structure **250** is illustrated as a transparent feature in FIG. **2**B in order to illustrate the features (such as the channel layers **120**) that the gate structure **250** covers. The gate structures **250** may be configured to extend lengthwise parallel to each other, for example, each along the Y-direction. In some embodiments, the gate structures **250** each wrap around the top surface and side surfaces of each of the fin structures **104**. In some embodiments, as described later, the gate structure **250** is first formed with a dummy gate stack of a different material, such as polysilicon, which is subsequently replaced with the gate dielectric layer **246** and the gate electrode layer **248** (and in some embodiments, the interfacial layer **242**). The dummy gate stacks **240** may be formed by a procedure including deposition, lithography, patterning, and etching processes. The deposition processes may include chemical vapor deposition (CVD) processes, atomic layer deposition (ALD) processes, physical vapor depositions (PVD), other suitable methods, or combinations thereof. The gate structure **250** further includes gate spacers. Gate spacers may include a single layer or a multi-layer structure. For example, in the depicted embodiment, a gate spacer layer **201** is formed over the top surface of the device, and a gate spacer layer **202** is formed over the gate spacer layer **201**. The gate spacer layers **201** and **202** may each include silicon nitride (Si.sub.3N.sub.4), silicon oxide (SiO.sub.2), silicon carbide (SiC), silicon oxycarbide (SiOC), silicon oxynitride (SiON), silicon oxycarbon nitride (SiOCN), carbon doped oxide, nitrogen doped oxide, porous oxide, or combinations thereof.

[0022] The transistor **100** further includes isolation features **150** within or over the substrate **102**, separating adjacent fin structures **104** from each other. The isolation features **150** may be shallow trench isolation (STI) features. In some examples, the formation of the isolation features **150** includes etching trenches into the substrate **102** between the active regions (the regions in which the fin structures are formed) and filling the trenches with one or more dielectric materials such as silicon oxide, silicon nitride, silicon oxynitride, other suitable materials, or combinations thereof. Any appropriate methods, such as a CVD process, an ALD process, a PVD process, a plasma-enhanced CVD (PECVD) process, a plasma-enhanced ALD (PEALD) process, and/or combinations thereof may be used for depositing the isolation features **150**. The isolation features

150 may have a multi-layer structure such as a thermal oxide liner layer over the substrate **102** and a filling layer (e.g., silicon nitride or silicon oxide) over the thermal oxide liner layer. Alternatively, the isolation features **150** may be formed using any other isolation technologies. As illustrated in FIG. **2**B, the fin structure **104** is located above the top surface of the isolation features **150**. In the depicted embodiment, the transistor **100** further includes inner spacers **206** between the gate structures **250** and the source/drain features **500**; contact etch stop layers **220** over the epitaxial source/drain features **500**; and interlayer dielectric (ILD) layer **230** over the epitaxial source/drain features **500** and over the contact etch stop layers **220**. FIGS. **2**B and **2**C have been abbreviated to provide a general picture of the transistor **100**, and do not include all details. Additional details of the gate structures **250** are described in conjunction with subsequent figures.

[0023] As described above, the transistor **100** is formed by replacing a dummy gate stack of the gate structure **250** with the gate dielectric layer **246** and the gate electrode layer **248** (and in some embodiments, the interfacial layer **242**). The disclosure below describes details for forming the gate dielectric layer **246**. Referring back to block **1010** of FIG. **1** and to FIG. **3**, a workpiece for the device **10** (or simply workpiece **10**) is received. FIG. **3** illustrates only a portion of the workpiece **10**, which is subsequently processed into the transistors **100** of FIGS. **2**B and **2**C (e.g. one of the transistors **100**A-**100**D and **100**A′-**100**D′). In other words, FIG. **3** illustrates the transistor **100** at an earlier processing stage than that in FIGS. **2**B and **2**C. More specifically, the transistor **100** at this processing stage includes all features described above with respect to FIGS. **2**B and **2**C with the exception that the gate structure **250** includes the dummy gate stack **240** rather than the gate dielectric layer **246**, the gate electrode layer **248** or the interfacial layer **242**. The dummy gate stack **240** may include any suitable materials, such as polysilicon. In some embodiments, the dummy gate stack **240** may include a multi-layer structure. For example, in some implementations, dummy gate stack may include a dummy gate dielectric layer and a dummy gate electrode layer.

[0024] Referring to FIG. **4**, the dummy gate stack **240** is selectively removed from the gate structure **250** to form openings. The etching process may be a dry etching process, a wet etching process, or combinations thereof. The etching process can be tuned, such that dummy gate stack **240** is removed without (or only minimally) etching other features of transistor **100**. Furthermore, following the removal of the dummy gate stack **240**, which exposes sidewall surfaces of the fin structures **104**, the remaining portions of the sacrificial semiconductor layers **110** are selectively removed to form additional openings. These openings collectively form gate trenches **241**. The gate trenches **241** expose portions of the channel layers **120** in 360° and further exposes the top surface of the substrate **102**.

[0025] FIG. **5** illustrates an expanded view of the transistor **100**, particularly, a portion of the gate trench **241** of the transistor **100**. Referring to FIG. **5**, the method **1000** proceeds to form an interfacial layer **242** in the gate trenches **241** and on the channel layers **120**. In some embodiments, interfacial layers **242** are formed on the exposed surfaces of the channel layers **120**. In some embodiments, the interfacial layers **242** improves the adhesion between the subsequently formed gate dielectric layer **246** and the channel layers **120**. In some embodiments, the interfacial layer **242** has a thickness in a range of about 5 Å to about 15 Å. In embodiments, the interfacial layer **242** includes a dielectric material, such as SiO.sub.2, HfSiO, SiON, other silicon-containing dielectric material, other suitable dielectric material, or combinations thereof. The interfacial layer **242** is formed by any suitable processes, such as thermal oxidation, chemical oxidation, ALD, CVD, other suitable process, or combinations thereof.

[0026] Referring to block **1020** of FIG. **1** and to FIG. **5**, a gate dielectric layer **246** is formed on the interfacial layer **242**, such as directly on and contacting (e.g. interfacing with) the interfacial layer **242**. Moreover, the gate dielectric layer **246** may be further formed over other surfaces exposed in the gate trenches **241**. The gate dielectric layer **246** may include a high-k dielectric material, such as HfO.sub.2, HfSiO, HfSiO.sub.4, HfSiON, HfLaO, HfTaO, HfTiO, HfZrO, HfAlO.sub.x, ZrO, ZrO.sub.2, ZrSiO.sub.2, AlO, AlSiO, Al.sub.2O.sub.3, TiO, TiO.sub.2, LaO, LaSiO, Ta.sub.2O.sub.3, Ta.sub.2O.sub.5, Y.sub.2O.sub.3, SrTiO.sub.3, BaZrO, BaTiO.sub.3 (BTO), (Ba,Sr)TiO.sub.3 (BST), hafnium dioxide-alumina (HfO.sub.2—Al.sub.2O.sub.3) alloy, other suitable high-k dielectric material, or combinations thereof. The gate dielectric layer **246** is formed by any of the processes described herein, such as ALD, CVD, PVD, oxidation-based deposition process, other suitable process, or combinations thereof. For example, a gate dielectric layer **246** may be conformally deposited over the interfacial layers **242** by an ALD process, such that gate dielectric layer **246** has a substantially uniform thickness and partially fills the gate trenches **241**. The gate dielectric layer **246** may be disposed on sidewall surfaces of the inner spacers **206**, as well as surrounding the channel layers **120**. In some embodiments, the gate dielectric layer **246** has a thickness of about 1 nm to about 3 nm. The gate dielectric layer **246** is formed surrounding the exposed portions of the channel layers **120** and reduces the size of the gate trenches **241**. The gate dielectric layer **246** separates the channel layers **120** and the gate electrode layer subsequently formed and is critical to determining threshold voltage of the transistor.

[0027] Referring to block **1030** of FIG. **1** and to FIG. **6**, a dipole layer **302** is deposited into the gate trenches **241**, or some of the gate trenches **241** as described later. As described above, the gate trenches **241** each surround portions of the channel layers **120** (e.g. the portions wrapped by the gate dielectric layer **246**) in 360° and also over a portion of the topmost channel layer **120** and a portion of the substrate **102**. Accordingly, the dipole layer **302** is formed on and surrounding the gate dielectric layer **246** such that the dipole layer **302** directly contacts the exposed surfaces of the gate dielectric layer **246**. The dipole layer **302** may be deposited by ALD, CVD, PVD, thermal oxidation, or other suitable methods, and may be deposited at a temperate in a range from about 100° C. to about 450° C. at a pressure in a range from about 1 torr to about 100 torr. The dipole layer **302** may include any suitable materials. In some embodiments, the dipole layer **302** may include an n-dipole material or a precursor to an n-dipole material. The n-dipole material may include germanium oxide (GeO.sub.2), yttrium oxide (Y.sub.2O.sub.3), lanthanum oxide (La.sub.2O.sub.3), strontium oxide (SrO), other suitable n-dipole materials, or combinations thereof. In some embodiments, the dipole layer **302** may include a p-dipole material or a precursor to a p-dipole material. The p-dipole material may include aluminum oxide (Al.sub.2O.sub.3), gallium oxide (Ga.sub.2O.sub.3), magnesium oxide (MgO), hafnium oxide (HfO.sub.2), titanium oxide (TiO.sub.2), zirconium oxide (ZrO.sub.2), zinc oxide (ZnO), other suitable p-dipole materials, or combinations thereof. As illustrated in FIG. **6**, the dipole layer **302** is spaced away from the channel layers **120** and from the gate dielectric layer **246** at this processing stage. As will be discussed, the dipole materials of the dipole layer **302** will be, in a subsequent step, thermally driven into the respective gate dielectric layer **246** such that they distribute across the gate dielectric layer **246** and around the interface between the gate dielectric layer **246** and the interfacial layer **242**. Accordingly, the dipole materials of the dipole layer **302** may thus be distributed closer or approach surfaces of the channel layers **120**. The n-dipole material configured this way in an NMOS serves to reduce the threshold voltage of the NMOS; while the n-dipole material configured this way in a PMOS serves to increase the threshold voltage of the PMOS. Similarly, the p-dipole material configured this way in a PMOS serves to reduce the threshold voltage of the PMOS; while the p-dipole material configured this way in an NMOS serves to increase the threshold voltage of the NMOS. Moreover, the amount and distribution of these dipole materials further affect the magnitude of change in the threshold voltages. Accordingly, threshold voltages of the transistors may be fine-tuned by simply engineering the presence or absence, chemical identity, as well as the distributions of the dipole materials within the gate dielectric layer **246**.

[0028] In some embodiments, the material of the dipole layer **302** can be designed based on the desired magnitude (or amount) of threshold voltage tuning. For example, using materials such as La.sub.2O.sub.3, Y.sub.2O.sub.3, or TiO.sub.2, the threshold voltage of the transistor **100** may be adjusted up (for p-type transistor) or down (for n-type transistor) in a range of about 20 mV to

about 450 mV. Moreover, the thickness of the dipole layer **302** may be further adjusted based on the desired magnitude of threshold voltage tuning. In some embodiment, a thicker dipole layer **302** generally allows (all things else being equal) more dipole material to enter the gate dielectric layer **246**, and leads to a greater change in the transistor's threshold voltage. In some embodiments, the dipole layer **302** may be deposited to a substantially uniform thickness in a range about 0.5 Å to about 10 Å in various embodiments, such as about 3 Å to about 5 Å. If the thickness is too small (such as less than 0.5 Å), the dipole layer **302** may be too weak for Vt tuning in some instances. If the thickness is too big (such as greater than 10 Å), the dipole layer **302** may be too strong for Vt tuning and may create side effects such as degraded mobility in the channel layers **120**.
[0029] Moreover, as described above, the device **10** includes a plurality of NMOS transistors and PMOS transistors. Each transistor may have device structures that resemble that of the transistor **100** described above with respect to FIGS. **2**B-**2**C, and **3-6**, with the exception that only a subset of the transistors include the dipole layer **302**. Accordingly, each transistor includes a respective gate trench **241** and may, where appropriate, receive the deposition of a respective portion of the dipole layer **302** on the respective portion of the gate dielectric layer. In other words, each of the transistors **100**A-**100**D and **100**A′-**100**D′ may include a portion **200**A-**200**D (corresponding to the portion **200** of FIG. **6**) including the gate dielectric layer **246** and the remaining portion of the gate trench **241**. Moreover, some of the portions **200**A-**200**D may include the dipole layer **302**, as described in detail later. In that regard, FIG. **7** illustrates portions **200**A-**200**D of the NMOS transistors **100**A-**100**D for the purpose of clearly illustrating aspects of the embodiments. While the portions **200**A-**200**D are depicted to be discontinuous from each other, it is understood that portions thereof may instead be connected to one or another. Moreover, although they are depicted side-by-side in a particular sequence, it is understood that any alternative relative position are contemplated by the present disclosure. Furthermore, although not explicitly illustrated below, transistors **100**A′-**100**D′ may include gate dielectric portions and dipole layer **302** similar to those NMOS transistors **100**A-**100**D, and undergo similar processing steps.
[0030] Referring to FIG. **7**, each of the portions **200**A-**200**D include a respective portion of the gate dielectric layer **246**, referred to as the gate dielectric portions **246**A-**246**D, respectively. The dipole layer **302** is shown here as a collection of circles representing chemical compositions of the dipole layer **302** of FIG. **6**. The number of the circles do not necessarily represent the concentrations or amounts of the chemical compositions. Although not explicitly depicted in FIG. **7**, the dipole layer **302** is further formed on sidewall surfaces and bottom surfaces of the gate dielectric portions **246**A-**246**D, similar to the dipole layer **302** formed on sidewall surfaces and bottom surfaces of the gate dielectric layer **246** of FIG. **6**. In the depicted embodiments, the dipole layer **302** is formed on surfaces of a subset, but not all, of the gate dielectric portions **246**A-**246**D. At this processing stage, the gate dielectric portions **246**B and **246**D may each have a thickness d**1**B and d**1**D, which may be about 1 nm to about 3 nm. In some embodiments, the thickness d**1**B and d**1**D may be similar to the thickness d**1**A and d**1**C of the respective gate dielectric portions **246**A and **246**C. If the thickness d**1**A-d**1**D is too small, such as less than about 1 nm, tunneling may increase drastically across the gate dielectric portions **246**A-**246**D thereby causing leakage. If the thickness d**1**A-d**1**D is too large, such as greater than about 3 nm, the gate capacitances may not be optimized.
[0031] Referring to block **1040** of FIG. **1** and to FIG. **8**, a process **402** is conducted to form a diffusion feature inside the gate dielectric layer **246** in regions configured for a subset of the transistors **100**A-**100**D. In some embodiments, the process **402** includes a thermal drive-in operation (alternatively referred to as an annealing process). In some embodiments, the dipole materials of the dipole layer **302** (e.g. dipole material La.sub.2O.sub.3), or dipole elements of the dipole layer **302** (e.g. the La and O elements of the dipole layer **302** having the dipole material La.sub.2O.sub.3) are at least partially driven into the gate dielectric layer **246**. For example, the thermal drive-in operation provides thermal energy to the workpiece **10** such that the mobility of the dipole material substantially increases, and the dipole elements diffuse into areas they interface

with (such as into the gate dielectric portions **246**B and **246**D interfacing with the dipole layer **302**). In an embodiment, the thermal drive-in operation is a soak anneal process at a temperature in a range from about 600° C. to about 1,000° C., such as about 700° C. to about 800° C., with O.sub.2, N.sub.2, or a mixture of O.sub.2 and N.sub.2 ambient. In another embodiment, the thermal drive-in operation is a furnace anneal process at a temperature in a range from about 300° C. to about 600° C. with O.sub.2, N.sub.2, or a mixture of O.sub.2 and N.sub.2 ambient for about 1 second to about 30 minutes. In yet another embodiment, the thermal drive-in operation is a spike anneal process. In still another embodiment, the thermal drive-in operation is a laser anneal process or a microwave anneal process at a temperature in a range from about 800° C. to about 1,200° C. with O.sub.2, N.sub.2, NH.sub.3, H.sub.2, or a mixture thereof for about 1 millisecond to about 10 seconds. The above ranges of temperature are selected such that the thermal drive-in operation does not adversely affect the existing structures and features of the device **10** and is yet sufficient to cause the dipole elements to migrate (or diffuse) from the dipole layer **302** into the gate dielectric layer **246** thereunder. In some embodiments, the dipole elements (e.g. La and O) diffuse into the gate dielectric layer **246** while maintaining their stoichiometric ratio as that in the dipole layer **302** (e.g. [La]:[O] at 2:3 for La.sub.2O.sub.3). In other words, the stoichiometric ratio of the dipole elements in the gate dielectric layer **246** (e.g. the ratio of the increase in La atomic concentration to the increase in O atomic concentration) is substantially the same as that of the dipole elements in the dipole layer **302**. In some other embodiments, the dipole elements diffuse into the gate dielectric layer **246** without maintaining the same stoichiometric ratio. Accordingly, the ratio of the dipole elements within the gate dielectric layer **246** (e.g. the ratio of the increase in La atomic concentration to the increase in O atomic concentration) may differ from that of the dipole layer **302**. As described later in detail, the chemical identity, the amount (or concentration), and the distributions of the dipole elements may be further adjusted to achieve the desired threshold voltage.

[0032] The thermal drive-in operation of process **402** causes a portion of the dipole materials of the layer **302** to diffuse into the gate dielectric portions **246**B and **246**D in a specific manner. As a result, the gate dielectric portions **246**B and **246**D each include a specific dipole material composition **2002**. Accordingly, the gate dielectric portions **246**B and **246**D may be said to include a diffusion feature of the dipole material composition **2002** (or simply diffusion feature **2002**). As will be discussed in more details below, the dipole material composition **2002** is configured differently from dipole material compositions in other gate dielectric portions, in chemical identities, concentrations, and/or distributions of the dipole materials (or dipole elements if the stoichiometric ratio is not maintained). This provides the tunability of the threshold voltages of the respective transistors separately from each other, and potentially, differently from each other. Although not explicitly illustrated in FIG. **8**, in some embodiments, the thermal drive-in operation of process **402** is configured to form an intermixing layer at the interface of the remaining portion of the dipole layer **302** and the modified gate dielectric portions **246**B and **246**D (e.g. modified with the dipole material composition **2002**). For example, the intermixing layers are formed on the modified gate dielectric portions **246**B and **246**D, which may each include the dipole material of dipole layer **302** at a concentration of about 40% to about 60%. In other words, this intermixing layer include characteristics (e.g. the etching resistances) of both the gate dielectric layer **246** and the dipole layer **302**. In some embodiments, the concentrations of the dipole material in the intermixing layers are each greater than the dipole material in the gate dielectric portions **246**B and **246**D, respectively. Meanwhile, the gate dielectric portions **246**A and **246**C do not include such intermixing layers.

[0033] As described above, a substantial portion of the dipole layer **302** are not driven into the gate dielectric portions **246**B and **246**D at the conclusion of the thermal drive-in operation of process **402**, and therefore remain on surfaces of the gate dielectric portions **246**B and **246**D (e.g. on surfaces of the intermixing layer on the surfaces of the gate dielectric portions **246**B and **246**D).

Still referring to block **1040** of FIG. **1** and to FIG. **8**, the process **402** proceeds to remove those remaining portions of the dipole layer **302** in an etching operation such that surfaces of the modified gate dielectric portions **246**B and **246**D (or doped gate dielectric portions **246**B and **246**D) are exposed. In some embodiments, the exposed surfaces of the modified gate dielectric portions **246**B and **246**D each includes the dipole material of the dipole layer **302** at a concentration less than about 40% to about 60%. The etching operation of the process **402** may include one or more etching processes (or stripping processes), which may be a dry etching process, a wet etching process, a reactive ion etching process, or another etching process and has a high etching selectivity with respect to the dipole layer **302** relative to the gate dielectric layer **246**. In some embodiments, the etching operation of the process **402** is a wet etching operation. It is noted that the removal of the remaining portions of the dipole layer **302** frees up limited spaces within the gate trenches **241** such that additional gate layers, such as another dipole layer (e.g. dipole layer **304** described later), another gate dielectric layer (such as gate dielectric layer **246**′ described later), other gate layers, or assistive layers that is required or beneficial for the proper functioning or improved functioning of the transistors, may be formed in the gate trenches **241**. Accordingly, embodiments implementing the etching operation of process **402** provide an improved device as compared to approaches not implementing such a process.
[0034] Following the process **402**, the gate dielectric portions **246**A-**246**D each has a thickness d**2**A-d**2**D, respectively. As described above, as the etching operation are configured to have an etch selectivity with respect to the dipole layer **302** relative to the gate dielectric layer **246** (or the unmodified gate dielectric portions **246**A and **246**C), the thicknesses d**2**A and d**2**C are each substantially the same as the thicknesses d**1**A and d**1**C. However, because of the presence of significant amount of dipole material, the etching operation may recess the intermixing layer on the modified gate dielectric portions **246**B and **246**D. Accordingly, following the etching operation, the thicknesses d**2**B and d**2**D of the gate dielectric portions **246**B and **246**D, respectively, are reduced as compared to the thicknesses d**1**B and d**1**D, and as compared to the thicknesses d**2**A and d**2**C. As will be described in detail later, the reduction in the thickness of the gate dielectric portions **246**B and **246**D reduces the channel resistance R.sub.ch of the transistors **100**B and **100**D, respectively. Moreover, as described later, the etching parameters of the etching operation of the process **402** may be adjusted to tune the amount of the intermixing layer removed, thereby tuning the remaining thickness d**2**B and d**2**D, and further to adjust the channel resistance R.sub.ch.
[0035] Following the process **402**, the gate dielectric portions **246**A and **246**C, as well as gate dielectric portions **246**B and **246**D are exposed in the gate trenches **241**. Moreover, the gate dielectric portions **246**B and **246**D may be thinner than the gate dielectric portions **246**A and **246**C. In other words, the surfaces of the gate dielectric portions in different substrate regions are unleveled (or stepped). Furthermore, the gate dielectric portions **246**B and **246**D now have compositions different from those of the gate dielectric portions **246**A and **246**C, in that they include the dipole material composition **2002** while the gate dielectric portions **246**A and **246**C do not. As described above, this distinction alone (e.g. the presence and absence of the dipole material composition **2002**) enables the transistors **100**B/**100**D to have threshold voltages that differ from the transistors **100**A/**100**C even when other aspects of the transistors are identical.
[0036] Referring to block **1050** of FIG. **1** and to FIG. **9**, the method proceeds to form another dipole layer in the gate dielectric layer in regions configured for certain transistors. For example, another dipole layer **304** is formed over a subset of the gate dielectric portions **246**A-**246**D. In some embodiments, the dipole layer **304** may similarly include an n-dipole material (e.g. GeO.sub.2, Y.sub.2O.sub.3, La.sub.2O.sub.3, SrO, other suitable n-dipole materials, or combinations thereof) or a precursor to an n-dipole material, a p-dipole material (e.g. Al.sub.2O.sub.3, Ga.sub.2O.sub.3, MgO, HfO.sub.2, TiO.sub.2, ZrO.sub.2, ZnO, other suitable p-dipole materials, or combinations thereof) or a precursor to a p-dipole material. In some embodiments, the dipole layer **304** includes a dipole material that is the same as that of the dipole

layer **302**. Having the dipole layers **302** and **304** with the same dipole material simplifies the processing and reduces costs. Alternatively, in some embodiments, the dipole layer **304** may include a dipole material different from the dipole layer **302**. Having the dipole layers **302** and **304** with different dipole materials provides further opportunities to adjust the threshold voltages of individual transistors, and potentially improves functionality. In some embodiments, the dipole layer **304** may be deposited to a substantially uniform thickness in a range from about 0.5 Å to about 10 Å in various embodiments, such as from about 3 Å to about 5 Å. If the thickness is too small (such as less than 0.5 Å), the dipole layer **304** may be too weak for Vt tuning in some instances. If the thickness is too large (such as greater than 10 Å), the dipole layer **304** may be too strong for Vt tuning and may create side effects such as degraded mobility in the channel layers **120**. In some embodiment, a thicker dipole layer **304** leads to a greater change in the transistor's threshold voltage. In some embodiments, the thickness of the dipole layer **304** may be less than the thickness of the dipole layer **302**. For example, in some embodiments, the dipole layer **302** may have a thickness t**1** and the dipole layer **304** may have a thickness t**2**. A difference between the thicknesses (t**1**−t**2**) may be about 0.1 Å to about 2 Å, for example, about 0.3 Å to about 1 Å. In some embodiments, this difference in thickness provides more latitude in adjusting the amount of the dipole material of the dipole layer **304** that subsequently diffuse into the gate dielectric portions **246**C and **246**D. If the difference is too small, there may not be sufficient difference in final dipole material compositions between the gate dielectric portions **246**C/**246**D from those of the gate dielectric portions **246**A/**246**B. If the difference is too large, the additional material may not bring significant difference in the diffusion behavior. Alternatively, in some embodiments, the thickness t**1** is less than the thickness t**2** to meet certain specific design needs (such as to create a specific threshold voltage cascade).

[0037] As described above, the dipole layer **304** is formed only on a subset of the gate dielectric portions **246**A-**246**D (in other words, only formed for a subset of the transistors **100**A-**100**D). In some embodiments, the dipole layer **304** is first formed across all transistors **100**A-**100**D, and subsequently removed from certain transistor regions, such as from the regions configured for the transistors **100**A and **100**B. Accordingly, the dipole layer **304** remain only on the gate dielectric portions **246**C and **246**D, and not on the gate dielectric portions **246**A and **246**B. The removal process may alter the thicknesses of some of the gate dielectric portions. Accordingly, at this processing stage, the gate dielectric portions may each have thicknesses d**3**A-d**3**D. In some embodiments, the thicknesses d**3**A and d**3**B may be similar to the thicknesses d**2**A and d**2**B (due to good etching selectivity); while the thicknesses d**3**C and d**3**D may be substantially the same as the thicknesses d**2**C and d**2**D. In the depicted embodiments, the dipole layer **304** directly interfaces with the gate dielectric portion **246**C without the dipole material composition **2002** formed therein. And the dipole layer **304** further directly interfaces with the gate dielectric portion **246**D with the dipole material composition **2002** formed therein. In other words, where the dipole layer **304** includes a material different from that of the dipole layer **302**, the top surface of the gate dielectric portion **246**D may include two different dipole materials.

[0038] Referring to block **1060** of FIG. **1** and to FIG. **10**, another process **404** is conducted. In some embodiments, the process **404** includes a thermal drive-in operation. The thermal drive-in operation of the process **404** may be similar to the thermal drive-in operation of the process **402** described above with respect to FIG. **8**. In some embodiments, the thermal drive-in operation of the process **404** may implement the same or different parameters from those of the thermal drive-in operation of the process **402**. In some embodiments, the thermal drive-in operation may implement a temperature lower than that of the thermal drive-in operation of the process **402**. For example, the thermal drive-in operation of the process **402** may implement a soak temperature T**1**, and the thermal drive-in operation of the process **404** may implement a soak temperature T**2**. A difference between the soak temperatures (T**1**−T**2**) may be about 50° C. to about 250° C., for example, about 100° C. to about 200° C. Adopting a higher annealing temperature during the process **402** and a lower annealing temperature during the process **404** allows better control of the diffusion behaviors of the dipole materials and consequently provides more precision control in the dipole material distribution in the respective gate dielectric portions. As a result, better tuning of the threshold voltages is received. If the difference in temperature (T**1**−T**2**) is too small, this benefit may be lost; while if the difference is too large, either the diffusion in the thermal drive-in operation may become hard to control, or the diffusion in the thermal drive-in operation may become insufficient. In some alternative embodiments, the temperature T**2** may be higher than the temperature T**1**. For example, in some embodiments, implementing a higher temperature T**2** may be useful to drive in dipole materials of the dipole layer **304** in an amount greater than that of the dipole layer **302** and/or with a distribution broader than that of the dipole layer **302**. In some embodiments, such greater amount and/or broader distribution may be beneficial to achieve particular threshold voltage configurations. Moreover, in some embodiments, the thermal drive-in operation of the process **404** may implement a time duration that is greater or less than that of the thermal drive-in operation of process **402**, such that the dipole material of the dipole layer **304** may migrate deeper or shallower into the gate dielectric portions **246**C and/or **246**D than those of the dipole layer **302** into the gate dielectric portions **246**B and/or **246**D. In some embodiments, the thermal drive-in operation of the process **404** is configured to drive the dipole materials of the dipole layer **304** deeper (than that of the process **402** on the dipole material composition **2002** at this processing stage) into the respective gate dielectric portions, despite using a lower anneal temperature and/or on a thinner dipole layer (as compared to the dipole layer **302**). In some embodiments, the thermal drive-in operation of the process **402** may have a time duration τ_1; the thermal drive-in operation of the process **404** may have a time duration τ_2. In some embodiments the time duration τ_2 may be greater than the time duration τ_1. For example, a difference ($\tau_2-\tau_1$) may be about 1 second to about 30 minutes, such as about 30 seconds to about 10 minutes. If the time difference is too small, the dipole material of the dipole layer **304** may not reach the desired depth; if the time difference is too large, the dipole material may diffuse so deep as to reach the channel layers **120** and adversely affect device performance.

[0039] The thermal drive-in operation of the process **404** has two primary effects. The first effect is that the dipole material compositions **2002** already within the gate dielectric portions **246**B and **246**D (see FIG. **10**) further migrate deeper therein, such that they may be distributed closer to the channel layers **120**. The second effect is that the dipole materials of the dipole layer **304** diffuse into the gate dielectric portions **246**C and the gate dielectric portions **246**D. As a result, the gate dielectric portions **246**C and **246**D both include a specific dipole material composition **2004**. The dipole material composition **2004** may be configured differently from the dipole material composition **2002** either in chemical identity or in concentrations. For example, the dipole material composition **2002** may include La_2O_3 while the dipole material composition **2004** may include Y_2O_3. Alternatively or additionally, the dipole material composition **2002** and **2004** may both include the respective dipole materials (either the same or different from each other) at different concentrations. These parameters may be adjusted to tune the threshold voltage differences between the transistors.

[0040] In the gate dielectric portion **246**D, the dipole material composition **2002** (see FIG. **8**) is modified with the further diffusion-in of the dipole materials from the dipole layer **304**, thereby forming dipole material composition **2042** (or diffusion feature **2042**). The dipole material composition **2042** may be similar to a combination of the dipole material composition **2002** and dipole material composition **2004**. Where the materials of the dipole layer **302** and dipole layer **304** are different, the dipole material composition **2042** includes a mixture in the gate dielectric portion **246**D. Where the materials of the dipole layer **302** and dipole layer **304** are the same, the dipole material composition **2002**, **2004**, and **2042** each include the same dipole material, although at different concentrations and with different distribution profiles. The dipole material composition **2042** may include the dipole material at the highest concentration which may be similar to the sum of the concentrations in the dipole material compositions **2002** and **2004**. Furthermore, in some embodiments, the dipole material composition **2002** may be distributed deeper than the dipole material composition **2004** (because it experienced two thermal drive-in operations as opposed to one for the dipole material composition **2004**); while the dipole material composition **2042** may have a broader distribution than the dipole material compositions **2002** and **2004**.

[0041] Still referring to block **1060** of FIG. **1** and to FIG. **10**, following the conclusion of the thermal drive-in operation, remaining portions of the dipole layer **304** on the gate dielectric layer are selectively removed in an etching operation of the process **404**, similar to the etching operation of the process **402** described above. Accordingly, valuable space of the gate trenches **241** are not occupied by the dipole layer **304**. Similar to those already described above with respect to FIG. **8**, the thermal drive-in operation of the process **404** may have been configured to produce an intermixing layer between the remaining portions of the dipole layer **304** and the gate dielectric portions **246**C and **246**D, which is absent from the gate dielectric portions **246**A and **246**B. Accordingly, the etching operation may be configured to have a good etching selectivity with respect to the dipole layer **304** relative to the gate dielectric layer **246**. Yet the presence of the intermixing layer may enable the etching operation to recess the top surfaces of the gate dielectric portions **246**C and **246**D. This reduces the thicknesses of the respective gate dielectric portions and the channel resistances in the transistors **100**C and **100**D. For example, following the etching operation of the process **404**, the gate dielectric portions **246**A-**246**D may each has a thickness d**4**A-d**4**D, respectively. The thickness d**4**A and d**4**B may be substantially the same as the thicknesses d**3**A and d**3**B due to the etching selectivity; while the thickness d**4**C and d**4**D may be less than the thicknesses d**3**C and d**3**D, respectively, due to the removal of the intermixing layer. In some embodiments, the thickness d**4**C may be configured to be less than the thickness d**4**B. This allows for a trend of declining thickness going from gate dielectric portions **246**A, **246**B, **246**C, to **246**D. Accordingly, as described later, a declining trend in the channel resistances is also received. This may be beneficial in application where continuous adjustment of resistances are important. The relative magnitude of the thickness d**4**C as compared to the thickness d**4**B may be controlled by controlling the parameters of the process **404** as compared to the process **402**. For example, the thermal drive-in operation of process **404** may be configured to be at a temperature higher than that of the process **402** or for a duration longer than that of the process **402**. Accordingly, the intermixing layer formed on the gate dielectric portion **246**C may be thicker than that on the gate dielectric portion **246**B. As a result, in the etching operation of the process **404**, a thicker portion of the gate dielectric portion **246**C (as compared to that of the gate dielectric portion **246**B in the etching operation of process **402**) may be removed. In some embodiments, the thickness d**4**B may be less than the thickness d**4**C. Such a configuration provides a zigzag trend in the channel resistances in the adjacent transistors **100**A-**100**D, and may be beneficial to control the overall resistance shift to stay within a certain threshold.

[0042] At this processing stage, gate dielectric portions **246**A-**246**D are each configured differently from one another. For example, the gate dielectric portion **246**A does not include a dipole material; the gate dielectric portion **246**B includes the dipole material composition **2002**; the gate dielectric portion **246**C includes the dipole material composition **2004**; and the gate dielectric portion **246**D includes the dipole material composition **2042** (or the combination of dipole materials **2002** and **2004**). The dipole material compositions **2002**, **2004**, and **2042** differ from each other in the chemical identity, concentration, and/or distribution within the respective gate dielectric portion. Because each of these parameters affect the threshold voltage of the transistor formed therefrom, transistors **100**A-**100**D incorporating these gate dielectric portions may provide different threshold voltages even if they include the same gate electrode layer. Moreover, the thickness d**4**A may be substantially similar to the thickness d**1**A, and may be greater than the thicknesses d**4**B-d**4**D; the thickness d**4**D may be less than the thicknesses d**4**B and d**4**C.

[0043] In some embodiments, referring to block **1070** of FIG. **1** and FIG. **11**, a process **406** is

conducted to further adjust the location and/or distribution of the dipole materials within the gate dielectric portions **246**A-**246**D. In some embodiments, the process **406** includes a thermal operation (or annealing operation). For example, the device **10** is annealed at a temperature of about 600° C. to about 1,000° C., such as about 700° C. to about 800° C., with O.sub.2, N.sub.2, or a mixture of O.sub.2 and N.sub.2 ambient. In another embodiment, the process **406** is a furnace anneal process at a temperature in a range from about 300° C. to about 600° C. with O.sub.2, N.sub.2, or a mixture of O.sub.2 and N.sub.2 ambient for about 1 second to about 30 minutes. In yet another embodiment, the process **406** is a spike anneal process. In still another embodiment, the process **406** is a laser anneal process or a microwave anneal process at a temperature in a range from about 800° C. to about 1,200° C. with O.sub.2, N.sub.2, NH.sub.3, H.sub.2, or a mixture thereof for about 1 millisecond to about 10 seconds. In some embodiments, the process **406** is configured to cause further migration of the dipole materials towards the channel layers **120**. As described later, this may improve the effectiveness of threshold voltage tuning. In some embodiments, the process **406** may be configured to cause the dipole materials within the gate dielectric portions to be randomized and reach a normalized distribution (as opposed to a trailing distribution), such that the control of the threshold voltage may be improved. In some embodiments, although not specifically depicted, the process **406** is instead conducted between the process **402** and the process **404**. In some embodiments, although not specifically depicted, the process **406** is conducted both between the process **402** and the process **404**, as well as following the process **404**. In some embodiments, the process **406** is omitted.
[0044] In some embodiments, additional dipole layers are formed over a subset of the gate dielectric portions, and additional thermal drive-in operations and etching operations are conducted to form transistors with additionally differentiated threshold voltages. In some embodiments, such additional dipole layers are omitted. In some embodiments (referring to the “A” branch of FIG. **1**), the method **1000** proceeds to form the gate electrode layer **248** (block **1080** of FIG. **1**) and to complete the fabrication of the gate structures **250**. However, in some other embodiments, referring to the “B” branch of FIG. **1** and to FIG. **12**, in some embodiments, another gate dielectric layer **246**′ is formed over the gate dielectric layer **246**. The gate dielectric layer **246**′ may be similar to the gate dielectric layer **246** described above. In some embodiments, the gate dielectric layer **246**′ may include the same or different material and experience the same or different processing from the gate dielectric layer **246**. For example, different portions of the gate dielectric layer **246**′ may be configured to include different dipole material compositions. In the depicted embodiments, the gate dielectric layer **246**′ are configured the same manner as the gate dielectric layer **246**. For example, the gate dielectric layer **246**′ of the transistor portion **200**A may include no dipole materials; the gate dielectric layer **246**′ of the transistor portion **200**B may include dipole material composition **2002**′ which may be similar to the dipole material composition **2002** described above with respect to FIG. **10**; the gate dielectric layer **246**′ of the transistor portion **200**C may include dipole material composition **2004**′ which may be similar to the dipole material composition **2004** described above with respect to FIG. **10**; the gate dielectric layer **246**′ of the transistor portion **200**D may include dipole material composition **2042**′ which may be similar to the dipole material composition **2042** described above with respect to FIG. **10**. Accordingly, after processing similar to those already described with respect to FIGS. **7-10**, the processed gate dielectric layer **246**′ may be similar or same as the processed gate dielectric layer **246** described above with respect to FIG. **10**. In other words, the processed gate dielectric layer **246** and **246**′ may be considered to be sublayers of a thicker gate dielectric structure. Additional gate dielectric layers may be further formed, and the cycle of processing of blocks **1020-1070** of FIG. **1** may be repeated until the desired total thickness of the gate dielectric layer is reached.
[0045] In some embodiments, forming the gate dielectric structure in a layer-by-layer approach may allow each layer (such as layer **246** and layer **246**′) to be thinner for better processability. Moreover, such an approach also allows better regulating of the distributions and profiles of the

dipole materials within the gate dielectric structure, thereby improving the tuning capacity and reliability of the threshold voltage. For example, the threshold voltages may be incrementally adjusted to provide better precision in control. Alternatively, the gate dielectric layer **246**′ may include a material that differs from the gate dielectric layer **246** and/or adopt a configuration of dipole materials different from that of the gate dielectric layer **246**. In such embodiments, more substrate regions with different dipole configurations may be provided which provides even more opportunity to create multi-Vt offerings. In some embodiments, additional gate dielectric layers may be formed over the gate dielectric layer **246**′ and processed in ways similar to those already described above with respect to FIGS. **7-11**.

[0046] Referring to block **1080** of FIG. **1** and to FIG. **12**, the method **1000** forms a gate electrode layer **248** over the processed gate dielectric layer(s). The gate electrode layer **248** is formed on the top surface of the gate dielectric layer **246**′ (or additional gate dielectric layers formed thereon). Referring back to FIGS. **2**B and **2**C, the gate electrode layer **248** wraps around the gate dielectric layer **246** over each of the channel layers **120**. The gate electrode layer **248** may completely or partially fill the gate trenches **241** in various embodiments. The gate electrode layer **248**, in combination with the gate dielectric layers, is designed to provide proper threshold voltages for the various NMOS and PMOS transistors. In some embodiments, the gate electrode layer **248** for the NMOS transistors **100**A-**100**D include any suitable n-type work function metal materials, such as titanium nitride (TiN), ruthenium (Ru), iridium (Ir), osmium (Os), rhodium (Rh), or combinations thereof. The gate electrode layer **248** for the PMOS transistors **100**A′-**100**D′ include any suitable p-type work function metal materials, such as titanium (Ti), aluminum (Al), tantalum (Ta), titanium aluminum (TiAl), titanium aluminum nitride (TiAlN), tantalum carbide (TaC), tantalum carbide nitride (TaCN), tantalum silicon nitride (TaSiN), or combinations thereof. Although not explicitly illustrated, the gate electrode layer **248** of NMOS and/or PMOS transistors may further include a fill metal layer. The fill metal layer may include any suitable materials, such as aluminum (Al), tungsten (W), copper (Cu), cobalt (Co), nickel (Ni), platinum (Pt), ruthenium (Ru), or combinations thereof. The conductive metal layer fills the remaining spaces of the gate trenches **241**. In some embodiments, a Chemical Mechanical Polishing operation is performed to expose a top surface of the ILD layer **230**. Accordingly, the gate dielectric layer **246** and the gate electrode layer **248** collectively form the high-k metal gate stack that replaces the original dummy gate stack **240**. The high-k metal gate stack and the gate spacer layers **201** and **202** collectively form the new gate structures **250**. The gate structures **250** engage multiple channel layers **120** to form multiple gate channels. Referring to block **1090** of FIG. **1**, the method **1000** proceeds to complete the fabrication of the device. For example, silicide features, contact features, via features, metal lines, passivation layers, among others may be formed.

[0047] In the present embodiment, the difference in the threshold voltages of the transistors can be completely tuned by the dipole material incorporations discussed above so that a common gate electrode layer **248** can be used for all the NMOS transistors (such as transistors **100**A-**100**D) or for all the PMOS transistors (such as transistors **100**A′-**100**D′). This obviates the need of using different work function metal layers (or different gate electrode layers) to achieve the variety in the threshold voltages. Thus, embodiments of the present disclosure enable the use of thinner work function metal layer(s) for the device **10** than other approaches, and are suitable for miniaturized multi-gate devices, such as nanosheet-based devices. It is noted that the gate electrode layer **248** may include multiple sub-layers, but it is still a common layer for multiple transistors (such as transistors **100**A-**100**D or transistors **100**A′-**100**D′). In some embodiments, the transistors **100**A-**100**D (or the transistors **100**A′-**100**D′) may each include different gate electrode layers to provide even further (or larger) tuning magnitude of the threshold voltages. In some embodiments, the present disclosure may further be implemented to enable transistors of different conductivity types (e.g. NMOS and PMOS) to use the same gate electrode layers.

[0048] The embodiments provided above allows effective tuning of the threshold voltages. FIG. **13** shows a chart illustrating the Vt tuning capability according to an embodiment of the method. In this embodiment, transistors in a device (such as the device **10**) are provided with **4** different threshold voltages. In this example, by incorporating dipole material composition **2002** in the transistor **100**B, the threshold voltage is adjusted by −40 mV as compared to the transistor **100**A without any dipole material. By incorporating dipole material composition **2004** in the transistor **100**C, the threshold voltage is adjusted by −76 mV as compared to the transistor **100**A. Moreover, by incorporating dipole material composition **2042** in the transistor **100**D, the threshold voltage is adjusted by −114 mV as compared to the transistor **100**A. Furthermore, these tuning capability matches well with the targeted values of the threshold voltages at −40 mV, −80 mV, and −120 mV, respectively. In other words, by simply tuning the presence and/or configuration (e.g. chemical identity, concentration, and distribution) of the dipole material compositions in the gate dielectric layers, multi-Vt device may be reliably achieved with tuning capacity as large as −120 mV. Moreover, the channel resistance penalty of an NMOS device formed this way has a reduced channel resistance penalty of about +0.03 kΩ-fin (as compared to +0.1 kΩ-fin with some other approaches). Meanwhile, the channel resistance penalty of a PMOS device formed this way has a negligible channel resistance penalty (as compared to +0.2 to +0.3 kΩ-fin with some other approaches). In other words, as compared to some other approaches, the method provides good Vt linearity, comparable leakage current (Igi), and low channel resistance penalties. Device improvements are thus received with the methods described herein.

[0049] Embodiments of the present disclosure may be adjusted to provide even wider threshold voltage tuning capacities, without substantial increase in processing complexities. For example, FIG. **14** provides a flow chart of an alternative embodiment of the method of the present disclosure (method **2000**) for fabricating a device **20**. FIG. **15** provides a plan view of an embodiment of devices (such as device **20**) of the present disclosure according to some embodiments of the present disclosure. FIGS. **16**-**21** provide cross-sectional views of the device **20** at different processing stages according to the flow chart of FIG. **14**. Method **2000** enables the fabrication of device **20** with NMOS transistors with six (6) different threshold voltages as well as PMOS transistors with another six (6) different threshold voltages. This is illustrated in FIG. **15**, the device **20** generally resembles the device **10** of FIG. **1** with the exception that it includes six NMOS transistors **100**A-**100**F and six PMOS transistors **100**A′-**100**F′. It is noted that reference numerals in subsequent descriptions have been reused for simplicity and clarity.

[0050] Referring to block **2020** of FIG. **14**, the method **2000** starts with receiving a workpiece having the various transistors formed thereon, each with channel layers engaged by a dummy gate stack, similar to that described above with respect to FIGS. **2**B and **2**C. The method **2000** proceeds to remove the dummy gate stack **240** thereby forming the gate trenches **241**, similar to that of FIG. **4**. Referring to block **2030** of FIG. **14** and to FIG. **16**, a gate dielectric layer **246** is formed in the gate trenches **241** around each of the channel layers **120**, similar to that described above with respect to FIG. **5**. Moreover, referring to block **2040** of FIG. **14** and to FIG. **16**, dipole layers are formed on each of the gate dielectric portions **246**A-**246**F. As illustrated in FIG. **16**, the dipole layers are differently configured in different regions **100**A-**100**F. For example, the gate dielectric portions **246**A and **246**B have no dipole layers formed thereon; the gate dielectric portions **246**C and **246**D have the dipole layer **304** thereon; and the gate dielectric portions **246**E and **246**F have both the dipole layer **302** and **304** thereon. Accordingly, at this processing stage, the surface of the transistor portions **100**A-**100**F have unleveled surfaces. In other words, the material layers surrounding the gate dielectric portions **246**E-**246**F in the transistor portions **200**E-**200**F are thicker than those surrounding the gate dielectric portions **246**C-**246**D in the transistor portions **200**C-**200**D, which are thicker than those surrounding the gate dielectric portions **246**A-**246**B in the transistor portions **200**A-**200**B. The dipole layer **302** may be similar to the dipole layer **302** described above, for example, including a same material and having a same thickness. The dipole layers **304** may include the same or different materials as the dipole layers **302**. Moreover, the dipole layers **302** and **304** may be conformal and have thicknesses adjusted based on the desired threshold voltage of the transistors, as described above. In the depicted embodiments, the dipole layer **304** has a thickness that is greater than the dipole layer **302** in order to provide greater concentration of dipole materials in the gate dielectric portions it interfaces with, as described in detail below.

[0051] Any suitable methods may be used to reach the configurations of the dipole layers **302** and **304** illustrated in FIG. **16**. For example, the dipole layer **302** may be first blanketly and/or conformally formed across all device regions **100**A-**100**F, and subsequently partially removed (e.g. patterned) such that it covers only the device regions **100**A, **100**B, **100**E, and **100**F. In other words, surfaces of the gate dielectric portions **246**C and **246**D in transistor portions **200**C and **200**D are exposed in the gate trenches **241**. The dipole layer **304** is then formed on all exposed surfaces, which includes, for example, on the surface of the dipole layer **302** on (or around) gate dielectric portions **246**A-**246**B and **246**E-**246**F, as well as on (or around) the surface of the gate dielectric portions **246**C and **246**D. Accordingly, the dipole layer **304** is spaced away from the gate dielectric layer **246** in transistor portions **200**A, **200**B, **200**E, and **200**F but directly contacts the gate dielectric layer **246** in transistor portions **200**C and **200**D. The dipole layers **302** and **304** in the device regions **100**A-**100**B is then removed while maintaining the dipole layers in the device regions **100**C-**100**F, for example, by implementing a mask element covering device regions **100**C-**100**F while exposing the device regions **100**A and **100**B. Any other suitable methods may alternatively be used.

[0052] Referring to block **2050** of FIG. **14** and to FIG. **17**, a process **502** is conducted. The process **502** may be similar to the process **402** described above with respect to FIG. **8**. For example, the process **502** includes a thermal drive-in operation that causes the dipole materials of the dipole layer **302**, as well as the dipole materials of the dipole layer **304**, to diffuse into the respective gate dielectric portions thereunderneath. The thermal drive-in operation may be similar to the thermal drive-in operations of the process **402**. For example, parameters of the thermal drive-in operation of process **502** may be adjusted to tune the amount of the dipole materials of the dipole layer **302** and/or the dipole materials of the dipole layer **304** to be driven into the gate dielectric layer **246**. The parameters may include the anneal temperature and time duration. These parameters, along with the chemical identities of the dipole materials of dipole layers **302** and **304**, as well as the thicknesses of the dipole layers **302** and **304**, are configured to provide proper dipole material compositions within the respective gate dielectric portions, similar to those already discussed with respect to device **10**. In some embodiments, because the dipole layer **302** directly interfaces with the gate dielectric portions **246**E and **246**F while the dipole layer **304** is spaced away from the gate dielectric portions **246**E and **246**F, a greater portion of the dipole layer **302** diffuses into the gate dielectric portions **246**E and **246**F than the dipole layer **304**. Moreover, because there are more dipole materials on the gate dielectric portions **246**E and **246**F (e.g. both dipole layers **302** and **304**) than on the gate dielectric portions **246**C and **246**D, a greater total amount (including dipole materials from the dipole layer **302** and the dipole layer **304**) diffuses into the gate dielectric portions **246**E and **246**F than into the gate dielectric portions **246**C and **246**D.

[0053] Following the thermal drive-in operation, the gate dielectric portions **246**A and **246**B do not include any dipole materials. The gate dielectric portions **246**C and **246**D include dipole materials from the dipole layer **304** but not from the dipole layer **302**. This combination of dipole material is referred to as the dipole material composition **2040**. The gate dielectric portions **246**E and **246**F includes dipole materials from the dipole layer **304**, as well as the dipole material from the dipole layer **302**. This combination of dipole material is referred to as the dipole material composition **2042**. Where the dipole layers **302** and **304** include different materials, the dipole material composition **2040** may be a single dipole material composition, while the dipole material **2042** includes a mixture. Where the dipole layers **302** and **304** include the same materials, the dipole material composition **2040** and the dipole materials composition **2042** may include the same material, although at different concentrations. Similar to those described above with respect to FIG. **8**, a portion of the dipole layer **302** and a portion of the dipole layer **304** remain on surfaces of the gate dielectric layer **246** at the conclusion of the thermal drive-in operation of process **502**. Subsequently, still referring to block **2050** of FIG. **14**, these remaining portions are removed in an etching operation of the process **502**, similar to the etching operation of the process **402** described above with respect to FIG. **8**.

[0054] Accordingly, at this processing stage, portions of the gate dielectric layer **246** in different substrate regions include different dipole material compositions. This distinction alone enables the device **20** to provide three (3) different threshold voltages for each transistor types. Additionally, similar to the process **402**, the thermal drive-in operation of the process **502** similarly produces an intermixing layer at the interface of the gate dielectric layer and the dipole layer **302** and/or **304**. In some embodiments, this intermixing layer is recessed or removed during the etching operation of the process **502**, similar to the situation in the etching operation of the process **402**. As a result, the thicknesses of the respective gate dielectric portions may be reduced. For example, following the process **502**, the gate dielectric portions **246**A-**246**F each has a respective thickness of d**3**A-d**3**F. The thickness d**3**A and d**3**B may be substantially the same as the thickness d**2**A and d**2**B (compare FIG. **16**) due to the etching selectivity implemented during the etching operation of the process **502**. However, the thickness d**3**C-d**3**F may be each less than the thickness d**2**C-d**2**F, respectively, due to the removal of the intermixing layer.

[0055] Referring to block **2060** of FIG. **14** and to FIG. **18**, another dipole layer **306** is formed on the gate dielectric portions **246**A-**246**F. In the depicted embodiments, the dipole layer **306** is formed on the gate dielectric portions **246**B, **246**D, and **246**F, but not on the gate dielectric portions **246**A, **246**C, and **246**E. The dipole layer **306** may be similar to the dipole layers **302** and/or **304**. For example, the dipole layer **306** may include a material that is similar to or different from those of the dipole layers **302** and/or **304**. In some embodiments, the dipole layer **306** has a thickness less than the dipole layer **304**, which in turn has a thickness less than the dipole layer **302**, for reasons similar to those discussed above with respect to FIG. **8**. Any suitable methods may be used to reach the configuration provided in FIG. **18**. For example, the dipole layer **306** may be formed conformally across all device regions **100**A-**100**F, and on surfaces of the modified gate dielectric layer **246** as seen in FIG. **17**. Subsequently, the dipole layer **306** is partially removed from surfaces of the gate dielectric portions **246**A, **246**C, and **246**E, for example, by implementing a patterning operation. The bottom surfaces of the remaining dipole layer **306** may interface with gate dielectric portions with different dipole material compositions. For example, the dipole layer **306** directly interfaces with the gate dielectric portion **246**B which includes no dipole materials; the dipole layer **306** directly interfaces with the gate dielectric portion **246**D which includes dipole material composition **2040**; and the dipole layer **306** directly interfaces with the gate dielectric portion **246**F which include dipole material composition **2042**.

[0056] Referring to block **2070** of FIG. **14** and to FIG. **19**, another process **504** is conducted. In some embodiments, the process **504** includes a thermal drive-in operation similar to the thermal drive-in operation of the process **404**. In some embodiments, the thermal drive-in operation may implement same or different parameters from those of the thermal drive-in operation of process **502**. In some embodiments, the thermal drive-in operation of the process **504** may implement a temperature lower than that of the thermal drive-in operation of the process **502**. For example, the thermal drive-in operation of the process **502** may implement a soak temperature T**1**, and the thermal drive-in operation of the process **504** may implement a soak temperature T**2**. A difference between the soak temperatures (T**1**−T**2**) may be about 50° C. to about 250° C., for example, about 100° C. to about 200° C. Adopting a higher annealing temperature during the process **502** and a lower annealing temperature during the process **504** allows better control of the diffusion behaviors of the dipole materials and subsequently provides more precision control in the dipole material distribution in the respective gate dielectric portions. Consequently, better tuning of the threshold

voltages is received. If the difference in temperature (T1−T2) is too small, this benefit may be lost; while if the difference is too large, either the diffusion in the thermal drive-in operation may become hard to control, or the diffusion in the thermal drive-in operation may become insufficient.
[0057] Furthermore, in some embodiments, the thermal drive-in operation of process **504** may implement a time duration that is greater or less than that of the thermal drive-in operation of process **502**, such that the dipole material of the dipole layer **306** may migrate deeper or shallower into the gate dielectric portions **246**B, **246**D, and/or **246**F than those of the dipole layer **302** or **304** into the gate dielectric portions **246**C and/or **246**E during the thermal drive-in operation of process **502** (see FIG. **17**). In some embodiments, the thermal drive-in operation of the process **502** may have a time duration τ.sub.1: the thermal drive-in operation of the process **504** may have a time duration τ.sub.2. In some embodiments the time duration τ.sub.2 may be greater than the time duration τ.sub.1. For example, a difference (τ.sub.2−τ.sub.1) may be about 1 second to about 30 minutes, such as about 30 seconds to about 10 minutes. In some embodiments, the thermal drive-in operation of the process **504** is configured to drive the dipole materials of the dipole layer **306** deeper (than the thermal drive-in operation of process **502**) into the respective gate dielectric portions, despite using a lower anneal temperature and/or on a thinner dipole layer (as compared to the dipole layer **302**). If the time difference is too small, the dipole material of the dipole layer **306** may not reach the desired depth; if the time difference is too large, the dipole material may diffuse so deep as to reach the channel layers **120** and adversely affect device performance.
[0058] Regardless, because the dipole materials compositions **2040** and **2042** in the gate dielectric portions **246**C and **246**E, respectively, each undergo two thermal drive-in operations, they may migrate to an overall deeper region of the gate dielectric portions and be closer to the channel layers **120**. Other parameters of the thermal drive-in operation of process **504** may be further adjusted based on the desired tuning range of the threshold voltages. As the conclusion of the thermal drive-in operation of process **504**, the gate dielectric portion **246**A includes no dipole materials; the gate dielectric portion **246**B includes the materials of the dipole layer **306** and not those from the dipole layers **302** or **304**. This dipole material combination is referred to as the dipole material composition **2600**. The gate dielectric portion **246**C is not affected by the formation of the dipole layer **306** or the thermal drive-in operation of process **504**, such that it maintains the dipole material composition **2040** as described above. The gate dielectric portion **246**D includes the dipole material composition **2040** prior to the formation of the dipole layer **306** and the thermal drive-in operation of process **504**, and further received diffusions of the materials from the dipole layer **306**. Accordingly, the gate dielectric portion **246**D includes materials from the original dipole layer **304** as well as from the dipole layer **306**. This combination is referred to as the dipole material composition **2640**. The gate dielectric portion **246**E is not affected by the formation of the dipole layer **306** or the thermal drive-in operation of process **504**, such that it maintains the dipole materials composition **2042** as described above. The gate dielectric portion **246**F includes the dipole material composition **2042** prior to the formation of the dipole layer **306** and the thermal drive-in operation of process **504**, and further received diffusions of the materials from the dipole layer **306**. Accordingly, the gate dielectric portion **246**F includes materials from the original dipole layers **302** and **304**, as well as from the dipole layer **306**. This combination is referred to as the dipole material composition **2642**.
[0059] Still referring to block **2070** of FIG. **14** and to FIG. **19**, following the completion of the thermal drive-in operation, another etching operation is conducted, similar to the etching operation of process **404** and/or process **502**. The etching operation of the process **504** not only removes the extra dipole materials over the gate dielectric layers, it also causes the thicknesses of the gate dielectric portions **246**A-**246**F to vary amongst themselves. For example, the gate dielectric portions **246**A-**246**F may have thicknesses d**4**A-d**4**F, respectively, following the etching operation of the process **504**. In some embodiments, the thickness d**4**A, d**4**C, and d**4**E remain substantially the same as the thickness d**3**A, d**3**C, and d**3**E, respectively; although the thickness d**4**B, d**4**D, and

d**4**F may be less than the thickness d**3**B, d**3**D, and d**3**F, respectively. As described above, this may be due to the formation of intermixing layers at the interface of the gate dielectric layer **246** and the dipole layer **306** in the relevant substrate regions. Also as described above, such reduction in layer thickness of the gate dielectric portions reduces the channel resistances R.sub.ch.

[0060] In some embodiments, the dipole layers **302**, **304**, and **306** are different from each other in chemical identities and/or layer thicknesses. Furthermore, as described above, parameters of the thermal drive-in operations of the processes **502** and **504** may be different from each other. Accordingly, the dipole materials **2600**, **2040**, **2640**, **2042**, and **2642** differ from each other in chemical identities, concentrations, and/or distributions within the respective portions of the gate dielectric layer **246**. As a result, when combined with the gate electrode layer, they each offer a unique threshold voltage that are different from each other and further different from that of the transistor **100**A which includes no such dipole materials, regardless whether the gate electrode layer is the same or different.

[0061] Referring to block **2080** of FIG. **14** and to FIG. **20**, a process **506** is conducted to further adjust the location and/or distribution of the dipole materials within the gate dielectric portions **246**A-**246**F. In some embodiments, the process **506** includes a thermal operation (or annealing operation). For example, the device **20** is annealed at a temperature of about 600° C. to about 1,000° C., such as about 700° C. to about 800° C., with O.sub.2, N.sub.2, or a mixture of O.sub.2 and N.sub.2 ambient. In another embodiment, the process **506** is a furnace anneal process at a temperature in a range from about 300° C. to about 600° C. with O.sub.2, N.sub.2, or a mixture of O.sub.2 and N.sub.2 ambient for about 1 second to about 30 minutes. In yet another embodiment, the process **506** is a spike anneal process. In still another embodiment, the process **506** is a laser anneal process or a microwave anneal process at a temperature in a range from about 800° C. to about 1,200° C. with O.sub.2, N.sub.2, NH.sub.3, H.sub.2, or a mixture thereof for about 1 millisecond to about 10 seconds. In some embodiments, the process **506** is configured to cause further migration of the dipole materials towards the channel layers **120**. As described later, this may improve the effectiveness of threshold voltage tuning. In some embodiments, the process **506** may be configured to cause the dipole materials within the gate dielectric portions to be randomized and reach a normalized distribution (as opposed to a trailing distribution), such that the control of the threshold voltage may be improved. In some embodiments, although not specifically depicted, the process **506** is instead conducted between the process **502** and the process **504**. In some embodiments, although not specifically depicted, the process **506** is conducted both between the process **502** and the process **504**, as well as following the process **504**. In some embodiments, the process **506** is omitted.

[0062] In some embodiments, additional dipole layers are formed over a subset of the gate dielectric portions, and additional thermal drive-in operations and etching operations are conducted to form transistors with additionally differentiated threshold voltages. In some embodiments, such additional dipole layers are omitted. Referring to branch “A” of FIG. **14**, in some embodiments, the method **2000** proceeds to block **2090** to form the gate electrode layer. In other embodiments, referring to branch “B” of FIG. **14**, the method **2000** proceeds back to the block **2030** to repeat the formation of another layer of gate dielectric layer **246**′. In such embodiments, referring to FIG. **21**, the additional gate dielectric layers **246**′ may be formed on top surfaces of the gate dielectric layer **246**, similar to that described above with respect to device **10**. Furthermore, referring to block **2090** of FIG. **14** and to FIG. **21**, gate electrode layer **248** is formed on top surface of the gate dielectric layer **246**′ (or additional gate dielectric layers formed thereon). In some embodiments, the portions of the gate electrode layer **248** interfacing with the gate dielectric portions **246**A-**246**F are identical, for example, having identical material compositions.

[0063] Referring to block **2100** of FIG. **14**, the method **2000** proceeds to complete the fabrication of the device. For example, silicide features, contact features, via features, metal lines, passivation layers, among others may be formed.

[0064] FIG. **22** shows a chart illustrating the Vt tuning capability according to an embodiment of the method. In this embodiment, transistors in a device (such as the device **20**) are provided with **6** different threshold voltages. In this example, by incorporating dipole material **2600** in the transistor **100**B, the threshold voltage is adjusted by −48 mV as compared to the transistor **100**A without any dipole material. By incorporating dipole material composition **2040** in the transistor **100**C, the threshold voltage is adjusted by −88 mV as compared to the transistor **100**A. Moreover, by incorporating dipole material composition **2640** in the transistor **100**D, the threshold voltage is adjusted by −121 mV as compared to the transistor **100**A; by incorporating dipole material composition **2042** in the transistor **100**E, the threshold voltage is adjusted by −174 mV as compared to the transistor **100**A; and by incorporating dipole material composition **2642** in the transistor **100**F, the threshold voltage is adjusted by −214 mV as compared to the transistor **100**A. Furthermore, these tuning capability matches well with the targeted values at −40 mV, −80 mV, −120 mV, −160 mV, and −200 mV respectively. In other words, by simply tuning the presence and/or configuration (e.g. chemical identity, concentration, and/or distribution) of the dipole materials in the gate dielectric layers, multi-Vt device may be reliably achieved with tuning capacity as large as −200 mV. Moreover, the channel resistance R.sub.ch for PMOS transistors by implementing the present disclosure, is reduced from about +0.2 kohm-fin to about 0.3 kohm-fin without implementing the methods provided herein to about +0.1 kohm-fin by implementing methods disclosed herein. As compared to some other approaches, the method provides good Vt linearity, comparable leakage current, and reduced channel resistance penalties, which all lead to device improvements.

[0065] FIGS. **23**A-**23**C offers more detail with regards to the diffusion and distribution of the dipole materials as results of the thermal drive-in operations **402**A (of process **402**) and **404**A (of process **404**), or **502**A (of process **502**) and **504**A (of process **504**). FIG. **23**A is a schematic view of the distribution of the dipole materials in the gate dielectric layer **246**. The line **702** depicts the interface between the gate dielectric layer **246** and the interfacial layer **242**; the line **708** represents the interface between the interfacial layer **242** and the channel layer **120**; the line **710** depicts the top surface of the gate dielectric layer **246** (which is also the interface between the gate dielectric layer and subsequently formed gate electrode layer **248**); the line **704** represents the center line **704** (or half-thickness line **704**) of the gate dielectric layer **246**; and the region **706** depicts a region refers to the region having a thickness of about 1 nm to about 2 nm centering on the interface **702**. In some embodiments, the dipole materials have dissimilar distributions in different transistors. For example, as seen in FIG. **23**A, the dipole materials of transistor **100**D has a wider distribution profile than the dipole materials of transistor **100**E. Specifically, there may be a larger amount of dipole materials distributed close to the top surface **710** of the gate dielectric portion **246**D than that of the gate dielectric portion **246**E; and there may be a smaller amount of dipole materials distributed within the region **706** of the gate dielectric portion **246**D than that of the gate dielectric portion **246**E. FIG. **23**B illustrates an example distribution of the dipole materials as a function of the location within the gate dielectric layer **246** and the interfacial layer **242**, as well as other adjacent layers. FIG. **23**B may have been simplified or conceptualized. The vertical axis of FIG. **23**B represent the dosage of the dipole materials as measured. In some embodiments, these thermal drive-in operations are designed so that some of the dipole materials can migrate through the gate dielectric layer **246** and reach the interfacial layer **242**. Moreover, although not explicitly depicted, in some embodiments, some dipole materials may further diffuse into the interfacial layer **242**. For example, at least a portion of the dipole materials are distributed in the region **706**. In some embodiments, having the dipole materials being close to the channel layer **120** allows greater tuning capability of the threshold voltage, even without increasing the amount of the dipole moment. However, having the dipole material too close to the channel layers may adversely affect other device characteristics. In some embodiments, dipole materials within the narrow region **706** has the maximum tuning efficacy of threshold voltages. In other words, threshold voltages are more sensitive to the dipole materials within the region **706** than to dipole materials outside the region **706**. For example, referring to FIG. **23**C, for a gate dielectric layer of HfO.sub.2 doped with La.sub.2O.sub.3 of the present disclosure, a linear fit of a plot of the change in threshold voltage (ΔV.sub.fb) to the ratio of the lanthanum element to hafnium element at the interface line **702** provides a coefficient of determination (R.sup.2) of about 0.98. Therefore, the threshold voltage responds not only to the chemical identities of the dipole materials, the concentrations of the dipole materials, as well as to the physical locations of the dipole materials in relationship to the interface **702**. Leveraging this relationship, additional tuning ability are achieved. It is noted that the present disclosure further offers tighter control over the ΔV.sub.fb as compared to other approaches not implementing disclosure herein. For example, a similar plot to that of FIG. **23**C not implementing the methods described here provides an R.sup.2 of less than 0.95.

[0066] Referring to FIGS. **23**A-**23**C and **24**, as described above, the parameters of the dipole layers **302**, **304**, and **306**, as well as the parameters of the thermal drive-in operations **502**A of process **502** and **504**A of process **504** may be adjusted to control the location and distribution of the diffused dipole materials. In some embodiments, the materials of the dipole layers **302**, **304**, and **306** are identical to each other. According, the dipole materials **2600**, **2040**, **2640**, **2042**, and **2642** are also identical, although in varying amounts. In some embodiments, it is desired to configure the adjacent transistors to have alternating (rather than continuously rising or declining) dipole material concentrations. For example, referring to FIG. **24**, it may be desirable to configure the concentration (or dosage) of the dipole materials to increase in the sequence of gate dielectric portions **246**A<**246**C<**246**B<**246**E<**246**D<**246**F. In some embodiments, this configuration maximized processing efficiency and control precision. Meanwhile, it may be desirable to configure the threshold voltages of the adjacent transistors to change continuously (rather than alternatingly). In some embodiments, this allows easier adjusting and selection of the threshold voltages during operation. In other words, it may be desirable to configure the threshold voltage tuning capacity to increase in the sequence of gate dielectric portions **246**A<**246**B<**246**C<**246**D<**246**E<**246**F. In some embodiments, this may be achieved by configuring distribution of the dipole materials in the gate dielectric portions **246**B, **246**D, and **246**F to be different from those in the gate dielectric portions **246**C and **246**E. For example, specifically referring to FIG. **23**B, the distribution of the dipole materials within the gate dielectric portions **246**C and **246**E are configured to peak (e.g. having the highest concentration) at or close to the interface line **702** (comparing the peak of the curve **246**C/**246**E relative to the line **702**). In some embodiments, the distribution of the dipole materials within the gate dielectric portions **246**C and **246**E arc configured to peak within the region **706**. Meanwhile, the distribution of the dipole materials within the gate dielectric portions **246**B, **246**D, and **246**F are configured to peak at, or close to, the center line **704** (or half-thickness line **704**) of the gate dielectric layer **246**. Moreover, the distribution of the dipole materials within the gate dielectric portions **246**A-**246**F are configured to have increasing concentrations at the interface line **702** according to the sequence **246**A<**246**B<**246**C<**246**D<**246**E<**246**F. Accordingly, because the threshold voltage is most sensitive to the dipole materials within the region **706** around the interface line **702**, the threshold voltage tuning capacity sequence is configured in the continuous fashion despite the overall concentration of the dipole material being configured in alternating fashion. In some embodiments, the above-described dipole material distribution is achieved by adopting a smaller layer thickness for dipole layer **306** than the dipole layer **304**, which has a smaller layer thickness than the dipole layer **302**, a lower anneal temperature (T**2**) for the thermal drive-in operation **504**A than that of the thermal drive-in operation **502**A (T**1**), or a longer anneal time duration (τ.sub.2) for the thermal drive-in operation **504**A than that of the thermal drive-in operation **502**A (τ.sub.1), as described above. In some embodiments (not depicted), the concentration of the dipole material in the gate dielectric portion **246**D is less than that in the gate dielectric portion **246**E.

[0067] As described above, the etching operations **402**B of process **402**, **404**B of process **404**,

502B of process **502**, and/or **504**B of process **504** remove a portion of the relevant gate dielectric portions due to presence of the intermixing layers. FIG. **25**A illustrates that following the etching operation **504**B of process **504**, which removes the remaining portions of dipole layer **306** on the gate dielectric portions **246**B, **246**D, and **246**F, as well as the intermixing layer thereon, the thickness of the gate dielectric portions are significantly reduced as compared to other approaches not implementing the methods of the present disclosure. For example, as compared to other approaches, the thickness of the gate dielectric portions **246**C and **246**E are each reduced by about 1 Å to about 2 Å. The reduction in the gate dielectric portion thickness leads to reduction in the channel resistances R.sub.ch. Referring to FIG. **25**B, as compared to those other approaches, embodiments of the present disclosure provide reduced R.sub.ch. For example, such reduction in the channel resistance of NMOS transistors may be about 0.1 kohm-fin to about 0.3 kohm-fin. Additionally, the reduced gate dielectric portions allow for greater processing window and easier fabrications.
[0068] In some embodiments, the etching parameters (such as etching temperature, etching solution concentration, etching time, other suitable wet etching parameters, or combinations thereof) of the various etching operations (**402**B of process **402**, **404**B of process **404**, **502**B of process **502**, and/or **504**B of process **504**) may be adjusted to provide different magnitude of thinning of the respective gate dielectric portions. FIGS. **26**A and **26**B illustrate an effect of etching time duration on the etching operation **504**B of process **504** as an example. In the depicted example, the etching condition 1 implements an etching time duration of 105 s; while the etching condition 2 implements an etching time duration of 210 s. As illustrated, the etching condition 2 leads to further reduced La dosage. Moreover, the channel resistances of the transistors with etching condition 2 are further reduced by about 0.03 kohm-fin to about 0.1 kohm-fin, such as about 0.05 kohm-fin to about 0.08 kohm-fin as compared to the etching condition 1.
[0069] Although not intended to be limiting, one or more embodiments of the present disclosure provide many benefits to a semiconductor device and the formation thereof. For example, the present disclosure provides methods of using dipole materials to engineer the transistors with threshold voltage tuning capabilities, in some instances without additional gate electrode layers (or work function metal layers). A threshold voltage tuning range of about 180 mV to about 220 mV is reached. This represents an improvement over other approaches by about 100 mV to about 150 mV. Moreover, because any additional dipole materials are removed after the thermal drive-in operation, no additional volume or space is required as compared to other approaches. In other words, this may be referred to as a “volumeless” approach. In some embodiments, by using methods provided herein, the need of patterning work function metal layer(s) is obliviated, making it very suitable for nanosized transistors and enabling continued downscaling. As compared to approaches that implement work function metals in adjusting threshold voltages, where the metals often remain in the finished devices, R.sub.ch penalty is improved. Moreover, in some embodiments, the large threshold voltage tuning capacity of the device **10** or **20** presented here allow fabricating of PMOS transistors with ultra-low threshold voltages, such as those conventionally only available with silicon germanium (SiGe) channel layers. Accordingly, in some embodiments, the present disclosure may be implemented in PMOS devices without SiGe channel layers, such as with Si channel layers alone. In some embodiments, the formation of SiGe channel layers require more complicated device processing. Accordingly, the present disclosure enables easier and more cost-effective fabrication of devices with similar threshold voltage tuning capabilities. In some other embodiments, the present disclosure may be implemented with the SiGe channel layers, so as to provide even greater threshold voltage tuning capacity. Furthermore, the present embodiments can be readily integrated into existing CMOS fabrication processes.
[0070] Methods **1000** and **2000** are merely examples and are not intended to limit the present disclosure to what is explicitly illustrated. For example, although the above disclosure describes the dielectric layers **246** being formed as part of the gate replacement process, in some alternative

embodiments, the gate dielectric layers **246** may be formed at an earlier processing stage, e.g. prior to the formation of the dummy gate stacks. Additional steps may be provided before, during, or after methods **1000** or **2000**, and some steps described can be replaced, eliminated, or moved around for additional embodiments of the method. Not all steps are described herein in detail for reasons of simplicity.

[0071] In one example aspect, the present disclosure is directed to a method. The method includes forming a dielectric layer on a semiconductor workpiece, forming a first patterned layer of a first dipole material on the dielectric layer, and performing a first thermal drive-in operation at a first temperature to form a diffusion feature in a first portion of the dielectric layer beneath the first patterned layer. The method also includes forming a second patterned layer of a second dipole material, where a first section of the second patterned layer is on the diffusion feature and a second section of the second patterned layer is offset from the diffusion feature. The method further includes performing a second thermal drive-in operation at a second temperature, where the second temperature is less than the first temperature. The method additionally includes forming a gate electrode layer on the dielectric layer.

[0072] In some embodiments, the first thermal drive-in operation is conducted for a first time duration, the second thermal drive-in operation is conducted for a second time duration, and the first time duration is less than the second time duration. In some embodiments, the first patterned layer has a first thickness, the second patterned layer has a second thickness, and the first thickness is greater than the second thickness. In some embodiments, the performing of the first thermal drive-in operation includes configuring the first thermal drive-in operation to form a first intermixing layer at an interface between the dielectric layer and first patterned layer. In some embodiments, the method further includes, after the performing of the first thermal drive-in operation, removing a remaining portion of the first patterned layer and the first intermixing layer. In some embodiments, the performing of the second thermal drive-in operation includes configuring the second thermal drive-in operation to form a second intermixing layer at an interface between the first section of second patterned layer and the dielectric layer, and a third intermixing layer at an interface between the second section of the second patterned layer and the dielectric layer. And the method further includes, after the performing of the second thermal drive-in operation, removing a remaining portion of the second patterned layer, the second intermixing layer, and the third intermixing layer. In some embodiments, the dielectric layer is a first dielectric layer, and the method further includes, after the performing of the second thermal drive-in operation, forming a second dielectric layer on the first dielectric layer, forming a third patterned layer of a third dipole material on the second dielectric layer, and performing a third thermal drive-in operation. In some embodiments, the forming of the first patterned layer includes forming a patterned sublayer covering a first region of the dielectric layer while exposing a second region of the dielectric layer, forming another sublayer on the patterned sublayer in the first region and on and interfacing with the dielectric layer in the second region, and collectively patterning the patterned sublayer and the another sublayer to expose the dielectric layer in a subset of the first region, thereby forming the first patterned layer. In some embodiments, the semiconductor workpiece includes a plurality of channel layers vertically stacked on a semiconductor substrate, and the dielectric layer is formed around the channel layers.

[0073] In one example aspect, the present disclosure is directed to a method. The method includes forming a dielectric layer on a semiconductor workpiece, forming a first patterned layer of a first dipole material on the dielectric layer, performing a first thermal drive-in operation for a first time duration, to drive a subset of the first dipole material into the dielectric layer, and performing a first etching operation to remove a remaining portion of the first dipole material. The method also includes forming a second patterned layer of a second dipole material on the dielectric layer, where a first portion of the second patterned layer is disposed on a first portion of the dielectric layer without the first dipole material and a second portion of the second patterned layer is disposed on a second portion of the dielectric layer with the first dipole material. The method further includes performing a second thermal drive-in operation for a second time duration, where the second time duration is greater than the first time duration. The method additionally includes performing a second etching operation to remove a remaining portion of the second dipole material, where the performing of the first etching operation includes recessing an intermixing region of the dielectric layer including the first dipole material.

[0074] In some embodiments, the method further includes forming a gate electrode layer on the dielectric layer, where the gate electrode layer and the dielectric layer surround a plurality of channel layers. In some embodiments, the performing of the second etching operation includes recessing an intermixing region of the dielectric layer that includes the second dipole material. In some embodiments, the performing of the second etching operation includes recessing an intermixing region of the dielectric layer that includes both the first dipole material and the second dipole material. In some embodiments, the performing of the first thermal drive-in operation includes performing at a first temperature, and the performing of the second thermal drive-in operation includes performing at a second temperature, and the first temperature is greater than the second temperature. In some embodiments, a difference between the first temperature and the second temperature is about 100° C. to about 200° C.

[0075] In one example aspect, the present disclosure is directed to a semiconductor device. The semiconductor device includes a semiconductor substrate, a first transistor on the semiconductor substrate and a second transistor on the semiconductor substrate. The first transistor includes a first channel, a first interfacial layer, a first gate dielectric layer on the first channel, and a first gate electrode layer on and interfacing with the first gate dielectric layer. The second transistor includes a second channel, a second interfacial layer, a second gate dielectric layer on the second channel, and a second gate electrode layer on and interfacing with the second gate dielectric layer. The first gate electrode layer and the second gate electrode layer have a same composition. The first gate dielectric layer includes a first dipole material composition that has a maximum concentration at a half-thickness line of the first gate dielectric layer. The second gate dielectric layer includes a second dipole material composition that has a maximum concentration at an interface between the second gate dielectric layer and the second interfacial layer.

[0076] In some embodiments, the dipole material includes one of germanium oxide (GeO.sub.2), yttrium oxide (Y.sub.2O.sub.3), Lanthanum oxide (La.sub.2O.sub.3), strontium oxide (SrO), magnesium oxide (MgO), hafnium oxide (HfO.sub.2), zirconium oxide (ZrO.sub.2), titanium oxide (TiO.sub.2), and aluminum oxide (Al.sub.2O.sub.3). In some embodiments, the second dipole material composition includes the first dipole material composition and a third dipole material composition. In some embodiments, the first channel is one of a first plurality of channels of the first transistor, and the second channel is one of a second plurality of channels of the second transistor. In some embodiments, the semiconductor device further includes a third transistor that has a third gate dielectric layer. The third gate dielectric layer is free of the first dipole material composition and free of the second dipole material composition.

[0077] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method, comprising: receiving a structure including a first region and a second region, wherein a portion of the first region overlaps with the second region; forming a dielectric layer over the first region and the second region; forming a first patterned layer of a first dipole material on the dielectric layer in the first region; performing a first thermal drive-in operation to drive the first dipole material into the dielectric layer; forming a second patterned layer of a second dipole material on the dielectric layer in the second region; performing a second thermal drive-in operation to drive the second dipole material into the dielectric layer; performing a thermal operation to adjust distribution of the first dipole material or both the first and the second dipole materials in the dielectric layer; and forming a gate electrode layer over the dielectric layer.
2. The method of claim 1, wherein the first dipole material and the second dipole material in the dielectric layer are different in chemical identity or in concentrations.
3. The method of claim 1, wherein performing the thermal operation is between performing the first thermal drive-in operation and performing the second thermal drive-in operation.
4. The method of claim 1, wherein performing the thermal operation is after performing the second thermal drive-in operation.
5. The method of claim 4, wherein the thermal operation is a first thermal operation, and wherein the method further comprises performing a second thermal operation after performing the first thermal drive-in operation and before performing the second thermal drive-in operation.
6. The method of claim 1, after performing the first thermal drive-in operation, further comprising selectively removing a remaining portion of the first patterned layer from the dielectric layer.
7. The method of claim 6, wherein selectively removing the remaining portion of the first patterned layer reduces a thickness of the dielectric layer in the first region.
8. The method of claim 1, after performing the second thermal drive-in operation, further comprising selectively removing a remaining portion of the second patterned layer from the dielectric layer, wherein selectively removing the remaining portion of the second patterned layer reduces a thickness of the dielectric layer in the second region.
9. A method, comprising: forming a first dielectric layer on a semiconductor structure; forming a first patterned layer of a first dipole material on the first dielectric layer in a region; forming a second patterned layer of a second dipole material on the first dielectric layer in the region; thereafter, performing a first thermal operation; forming a second dielectric layer on the first dielectric layer; forming a third patterned layer of the first dipole material on the second dielectric layer in the region; forming a fourth patterned layer of the second dipole material on the second dielectric layer in the region; thereafter, performing a second thermal operation; and forming a gate electrode layer on the second dielectric layer.
10. The method of claim 9, further comprising: performing a third thermal operation before forming the second patterned layer; and performing a fourth thermal operation before forming the fourth patterned layer.
11. The method of claim 9, further comprising: performing a third thermal operation after performing the first thermal operation and before forming the second dielectric layer; and performing a fourth thermal operation after performing the second thermal operation.
12. The method of claim 9, before forming the second patterned layer, further comprising forming a fifth patterned layer of a third dipole material on the first dielectric layer in the region.
13. The method of claim 9, wherein performing the first thermal operation drives a first portion of the second dipole material from the second patterned layer into the first dielectric layer, and wherein performing the second thermal operation drives a second portion of the second dipole material from the fourth patterned layer into the second dielectric layer.
14. The method of claim 9, wherein the region is a first region, wherein forming the first patterned layer further comprises forming the first patterned layer in a second region, and wherein forming the second patterned layer does not form the second patterned layer in the second region.

15. A method, comprising: forming a dielectric layer on a channel member of a semiconductor structure; forming a first layer of a first dipole material on the dielectric layer; performing a first thermal drive-in operation to drive the first dipole material into the dielectric layer; forming a second layer of a second dipole material on the dielectric layer; performing a second thermal drive-in operation to drive the second dipole material into the dielectric layer; performing a thermal operation, resulting in normalized distribution of the first dipole material or both the first and the second dipole materials; and forming a gate electrode layer over the dielectric layer.
16. The method of claim 15, before performing the first thermal drive-in operation, further comprising forming a third layer of a third dipole material on the dielectric layer.
17. The method of claim 16, wherein performing the first thermal drive-in operation drives the third dipole material into the dielectric layer.
18. The method of claim 15, wherein during the second thermal drive-in operation, the first dipole material migrates closer to the channel member.
19. The method of claim 15, wherein performing the thermal operation is after performing the second thermal drive-in operation.
20. The method of claim 19, wherein the thermal operation is a first thermal operation, wherein the method further comprises performing a second thermal operation before performing the second thermal drive-in operation and after performing the first thermal drive-in operation.